VERONIKA YAKUBU

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PROFESSIONAL SUMMARY

I am a Computational Engineer with extensive industrial experience in CFD simulation, thermal-fluid analysis and predictive simulation across various applications. I apply advanced numerical methods and critical engineering thinking to develop and optimize engineering solutions throughout the product lifecycle. My experience spans the complete simulation workflow, from geometry preparation and pre-processing through numerical model definition and validation against experimental data to post-processing and technical reporting. I have been responsible for design evaluation and approval based on predictive simulation results, balancing performance, efficiency and safety. I also have extensive experience improving simulation methodologies through automation, scripting, parallel computing and process optimization. I communicate complex technical findings through clear visualizations and concise reports, enabling effective engineering decisions within multidisciplinary teams.

PROFESSIONAL EXPERIENCE

Career break (full time parenting)

Jun 2025 – Present

BMW Group via IQ Intelligentes Ingenieur Management GmbH

Jul 2023 – May 2025 | Munich

Simulation Engineer CFD - Brake System Longitudinal Dynamics

- Approved brake system designs across multiple BMW and Rolls-Royce vehicle models in several development phases, covering ICE, hybrid, and electric powertrains
- Optimized and validated CFD models for airflow, heat transfer, and thermal management, ensuring compliance with industry-standard safety requirements and accurate representation of experimental test results
- Standardized and automated simulation workflows and CAE methodology through scripting and macros, improving engineering efficiency and reproducibility
- Steered technical decisions across Design, Construction, Vehicle Integration and Product Management teams by aligning cross-functionally and presenting simulation findings and technical reports throughout the development cycle
- Coached an engineering student in design optimization techniques using STARCCM+ and Shapemodule
- Owned a cross-functional initiative uniting 3D CFD and FEA into a single fluid-solid predictive deformation workflow in STAR-CCM+, partnering with testing and product management to embed the method in development

Siemens Industry Software GmbH

Jul 2022 – Apr 2023 | Munich

Working Student CFD Simulation Turbomachinery

- Implemented a complex, cutting-edge simulation model with extreme physical conditions to expand the company portfolio
- Thesis work titled: Numerical modeling of aero performance for the Energy Efficient Engine High-pressure Compressor

AVL Deutschland GmbH

Feb 2022 – Jun 2022 | Munich

Working student CFD Simulation

- Validated a new advanced adaptive meshing algorithm on benchmark cases, approving its release for production use
- Documentation and technical report generation

BMW Group

Aug 2021 – Dec 2021 | Munich

Internship CFD Simulation Thermodynamics

- Pre-processing, simulation, evaluation and validation of IC engines' air intake system against experimental data
- Optimized and automated pre- and post-processing workflows, reducing manual effort across simulation setups

AVL Hungary Kft.

Jul 2018 – Nov 2020 | Budapest, Hungary

Simulation Engineer CFD Automotive

- Performed steady-state CFD simulations of thermal management systems for ICE, EV and HEV, covering engine water jacket cooling, cooling plate design optimization and immersion cooling analysis for battery systems
- Built training materials and served as the go-to technical advisor for battery cooling simulation, standardizing knowledge across a team of 15 engineers

EDUCATION

Technical University of Munich *Apr 2023*
Master's Degree in Computational Science and Engineering

University of Miskolc, Hungary *Jul 2018*
Bachelor's Degree in Materials Engineering

TECHNICAL SKILLS

General: MS Office, Linux, Windows, Git, VS Code, SAP

Simulation Software: STAR-CCM+, OpenFOAM, AVL FIRE, Abaqus, ANSYS Mechanical, Simulink

Pre/Post Processing: ANSA, ParaView, META, CATIA V5

Programming & HPC: Python, bash, Matlab, C++, Java, MPI/OpenMP, CUDA

CORE STRENGTHS

Computational Engineering | CFD | FEA | FSI | CAD | HPC | CAE Automation | Simulation-Driven Product Development | Automotive | Design Optimization | Thermal Management | Conjugate Heat Transfer | Multiphysics Simulation | Numerical Methods | Applied Mathematics | Parallel Computing | Validation | Data Analysis | Materials Engineering

PROJECTS & PAPERS

Numerical Modeling of Aero Performance for the Energy Efficient Engine High-Pressure Compressor (Master's Thesis) *Apr 2023*

- Developed a high-fidelity aero performance model of the Energy Efficient Engine High-Pressure Compressor in STAR-CCM+, achieving a converging simulation with realistic pressure drop across 6 compressor stages

Fluid-Structure Interaction - Unit testing and simulation with the generalized α-method in Kratos Multiphysics for FSI application *Sep 2022*

- Fluid-Structure Interaction Practical exercise TUM CSE

Coupled numerical simulation of a simple battery cooling system using OpenFOAM and preCICE *Dec 2021*

- Validated the OpenFOAM–preCICE coupling workflow by building a simple, reproducible battery cooling model, achieving balanced temperatures and demonstrating successful multiphysics coupling

Building a 2D numerical solver in C++ using relaxation methods, conjugate gradient and multigrid algorithm *Jul 2021*

- Solver programming Exercice TUM CSE

A numerical solution to the 2D incompressible Navier-Stokes equation using staggered grid *Feb 2021*

- Practical Numerical Implementation TUM CSE

CFD modeling of FOXBG biomass gasifier using OpenFOAM (Bachelor's Thesis) *Jun 2018*

- Industry process optimization for the RISE Research Institute

LANGUAGES

English (C1), German (C1), Hungarian (native)